

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – V

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Part B - Program

a) Create a web application to demonstrate the use of different types of Cookies.

Algorithm:

STEP 1 :- START

STEP 2 :- Initialize the ViewState counter and display/increment it when the View State button is clicked.

STEP 3 :- Store the entered name in a cookie, redirect to the second page, and display the cookie value.

STEP 4 :- STOP

Code:

Webform1.aspx :-

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="Practical_5C.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>ViewState and Cookies</title>
</head>
<body>
<form id="form1" runat="server">
<asp:Label ID="Label1" runat="server"></asp:Label>
<br /><br />
<asp:Button ID="Button1" runat="server"
Text="View State"
OnClick="Button1_Click" />
<br /><br />
Enter Name :
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
<br /><br />
<asp:Button ID="Button2" runat="server"
Text="Cookies"
OnClick="Button2_Click" />
</form>
</body>
```

</html>

Webform1.aspx.cs :-

```
using System;
using System.Web;
namespace Practical_5C
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                ViewState["count"] = 1;
            }
        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            int count = Convert.ToInt32(ViewState["count"]);
            Label1.Text = "View State Count = " + count;
            count++;
            ViewState["count"] = count;
        }
        protected void Button2_Click(object sender, EventArgs e)
        {
            HttpCookie c = new HttpCookie("name");
            c.Value = TextBox1.Text;
            Response.Cookies.Add(c);
            Response.Redirect("WebForm2.aspx");
        }
    }
}
```

Webform2.aspx :-

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"
Inherits="Practical_5C.WebForm2" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title>Cookie Demo</title>
</head>
<body>
    <form id="form1" runat="server">
        <asp:Label ID="Label1" runat="server"></asp:Label>
    </form>
</body>
</html>
```

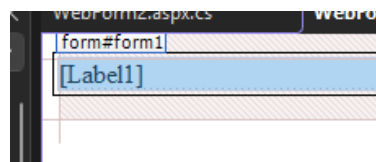
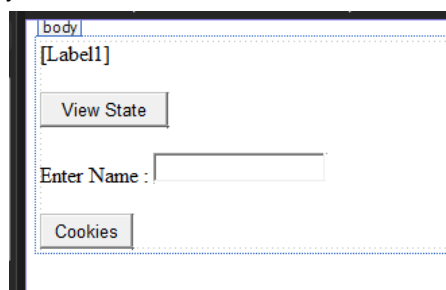
Webform2.aspx.cs :-

```
using System;
namespace Practical_5C
```

```

{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Request.Cookies["name"] != null)
            {
                Label1.Text = "Welcome " + Request.Cookies["name"].Value;
            }
            else
            {
                Label1.Text = "Cookie Not Found";
            }
        }
    }
}

```



OUTPUT :-



b) Create a Product Table in SQL Server and perform Insert Operation using Entity Framework Core.

Algorithm:

STEP 1 :- START

STEP 2 :- Establish a connection to the SQL Server database using Entity Framework Core.

STEP 3 :- Create a Product object and insert it into the Products table.

STEP 4 :- Save the changes and display the message "Product Inserted Successfully".

STEP 5 :- STOP

Code:

HomeController.cs :-

using Microsoft.AspNetCore.Mvc;

```
public class HomeController : Controller
{
    private readonly AppDbContext _context;
    public HomeController(AppDbContext context)
    {
        _context = context;
    }
    public IActionResult Index()
    {
        Product p = new Product();
        p.Name = "Saqlain";
        p.Salary = 50000;
        _context.Products.Add(p);
        _context.SaveChanges();
        return Content("Product Inserted Successfully");
    }
}
```

AppDbContext.cs :-

```
using Microsoft.EntityFrameworkCore;
using Practical_5B.Models;
public class AppDbContext : DbContext
{
    public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
    {
    }
    public DbSet<Product> Products { get; set; }
}
```

Program.cs :-

```
using Microsoft.EntityFrameworkCore;
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllersWithViews();
builder.Services.AddDbContext<AppDbContext>(options =>
options.UseSqlServer(
builder.Configuration.GetConnectionString("DefaultConnection")));
var app = builder.Build();
app.UseStaticFiles();
app.UseRouting();
app.MapControllerRoute(
name: "default",
pattern: "{controller=Home}/{action=Index}/{id?}");
app.Run();
```

Product.cs :-

```
public class Product
{
    public int Id { get; set; }
    public string Name { get; set; }
    public int Price { get; set; }
```

```
}
```

ErrorViewModel.cs :-

```
namespace Practical_5B.Models
```

```
{
```

```
    public class ErrorViewModel
```

```
    {
```

```
        public string? RequestId { get; set; }
```

```
        public bool ShowRequestId => !string.IsNullOrEmpty(RequestId);
```

```
    }
```

```
}
```

appsettings.json :-

```
{
```

```
    "ConnectionStrings": {
```

```
        "DefaultConnection":
```

```
        "Server=(localdb)\\MSSQLLocalDB;Database=ProductDB;Trusted_Connection=True;TrustServerCertificate=True;"
```

```
    }
```

```
}
```

Output:

```
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:7224
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5091
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\mvluc\source\repos\Practical_5B\Practical_5B
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (56ms) [Parameters=[@p0='?' (Size = 4000), @p1='?' (DbType = Int32)], CommandType='Text', CommandTimeout='30']
      SET IMPLICIT_TRANSACTIONS OFF;
      SET NOCOUNT ON;
      INSERT INTO [Products] ([Name], [Price])
      OUTPUT INSERTED.[Id]
      VALUES (@p0, @p1);
```

Product Inserted Successfully

```
22
23 SELECT * FROM Products;
24 Saqlain Khan T013
```

100 % 1 0

T-SQL Results Message

	Id	Name	Price
1	1	Laptop	50000
2	2	Laptop	50000
3	3	Laptop	50000

- c) Develop a web application to perform CRUD Operations (Create, Read, Update, Delete) on the Employee table using EF Core.

Algorithm:**STEP 1 :-** START**STEP 2 :-** Connect to the SQL Server database.**STEP 3 :-** Perform CRUD operations (Create, Read, Update, Delete) on Employee records.**STEP 4 :-** Display the updated Employee information to the user.**STEP 5 :-** STOP**Code:****HomeController.cs :-**

using System.Diagnostics;

using Microsoft.AspNetCore.Mvc;

using Practical5C.Models;

namespace Practical5C.Controllers

```
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return View();
        }
        public IActionResult Privacy()
        {
            return View();
        }
        [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]
        public IActionResult Error()
        {
            return View(new ErrorViewModel { RequestId = Activity.Current?.Id ??
HttpContext.TraceIdentifier });
        }
    }
}
```

appsettings.json :-

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "MyConnection":
"Server=(localdb)\\MSSQLLocalDB;Database=EmployeeDB;Trusted_Connection=True;TrustServer
Certificate=True",
  }
}
```

```

    "AppDbContext":
    "Server=(localdb)\\mssqllocaldb;Database=AppDbContext;Trusted_Connection=True;MultipleActiveResultSets=true"
    }
    }

```

Program.cs :-

```

using Microsoft.EntityFrameworkCore;
using Practical5C.Data;
var builder = WebApplication.CreateBuilder(args);
// Add services to the container.
builder.Services.AddControllersWithViews();
builder.Services.AddDbContext<AppDbContext>(options =>
options.UseSqlServer(builder.Configuration.GetConnectionString("MyConnection")));

var app = builder.Build();
// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    // The default HSTS value is 30 days. You may want to change this for production scenarios, see
    https://aka.ms/aspnetcore-hsts.
    app.UseHsts();
}
app.UseHttpsRedirection();
app.UseRouting();
app.UseAuthorization();
app.MapStaticAssets();
app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}")
    .WithStaticAssets();
app.Run();

```

AppContextDb.cs :-

```

using Microsoft.EntityFrameworkCore;
using Practical5C.Models;
namespace Practical5C.Data
{
    public class AppDbContext:DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }
        public DbSet<Employee> Employee { get; set; }
    }
}

```

Output:

Index

[Create New](#)

Name	Department	Salary	
Saqlain Khan	IT	50000	Edit Details Delete

Create

Employee

Name

Reze

Department

Heart

Salary

1000000000000000000

Create

[Back to List](#)

Delete

Are you sure you want to delete this?

Practical5C.Models.Employee

Name	Reze
Department	Heart
Salary	1E+18

Delete

[Back to List](#)

Index

[Create New](#)

Name	Department	Salary	
Saqlain Khan	IT	50000	Edit Details Delete

d) Authentication and Authorization in ASP.NET Core.

Algorithm:

STEP 1 :- START

STEP 2 :- Accept the username and password from the user.

STEP 3 :- Verify the entered credentials and authenticate the user using cookie authentication.

STEP 4 :- Grant access to the secure page if login is successful; otherwise, display an error message.

STEP 5 :- STOP

Code:

HomeController.cs :-

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Authentication;
using Microsoft.AspNetCore.Authentication.Cookies;
using System.Security.Claims;
namespace AuthDemo.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return View();
        }

        [HttpPost]
        public async Task<IActionResult> Index(string username, string password)
        {
            if (username == "Himani" && password == "123")
            {
                var claims = new List<Claim>
                {
                    new Claim(ClaimTypes.Name, username)
                };

                var identity = new ClaimsIdentity(claims,
                    CookieAuthenticationDefaults.AuthenticationScheme);

                await HttpContext.SignInAsync(
                    CookieAuthenticationDefaults.AuthenticationScheme,
                    new ClaimsPrincipal(identity));

                return RedirectToAction("Secure");
            }

            ViewBag.Message = "Invalid Username or Password";
            return View();
        }

        [Authorize]
        public IActionResult Secure()
        {
            return Content("Welcome! Login Successful.");
        }

        public async Task<IActionResult> Logout()
        {
            await HttpContext.SignOutAsync(
                CookieAuthenticationDefaults.AuthenticationScheme);

            return RedirectToAction("Index");
        }
    }
}
```

```
}  
}
```

Index.cshtml :-

```
<h2>Login</h2>  
<form method="post">  
  Username :  
  <input type="text" name="username" /><br /><br />  
  Password :  
  <input type="password" name="password" /><br /><br />  
  <input type="submit" value="Login" />  
</form>  
<p style="color:red">  
  @ViewBag.Message  
</p>
```

Program.cs :-

```
using Microsoft.AspNetCore.Authentication.Cookies;  
  
var builder = WebApplication.CreateBuilder(args);  
builder.Services.AddControllersWithViews();  
builder.Services.AddAuthentication(CookieAuthenticationDefaults.AuthenticationScheme)  
  .AddCookie(options =>  
  {  
    options.LoginPath = "/Home/Index";  
  });  
builder.Services.AddAuthorization();  
var app = builder.Build();  
app.UseStaticFiles();  
app.UseRouting();  
app.UseAuthentication();  
app.UseAuthorization();  
app.MapControllerRoute(  
  name: "default",  
  pattern: "{controller=Home}/{action=Index}/{id?}");  
app.Run();
```

Output:

Practical_5D Home Privacy

Login

Username :

Password :

Practical_5D [Home](#) [Privacy](#)

Login

Username :

Password :

Welcome! Login Successful.